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VALIDATION OF KINEMATIC HARDENING FOR MATERIAL USING LOW CYCLE FATIGUE

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ABSTRACT

This study investigates the low-cycle fatigue (LCF) behavior of SA508 Gr.3 steel using finite element analysis (FEA). The experimental methods referenced were sourced from prior research on strain-controlled fatigue tests [1], while this work focuses on validating the FEA conducted in ABAQUS. Material properties utilized were extracted from ASME Section II Part D M, 2023 [2]. Results indicate that the proposed FEA model aligns well with experimental data, offering reliable predictions within 85% of observed values. The study highlights the importance of LCF in designing durable components in various industrial applications.

Keywords: Low-cycle fatigue (LCF), Finite element analysis (FEA), Kinematic hardening

NOMENCLATURE

Place nomenclature section, if needed, here. Nomenclature should be given in a column, like this:

VLCF	Very Low Cycle Fatigue
PRD	Table for Poisson's Ratio & Density
Table U	Table for Ultimate tensile strength
Table TM-1	Table for Modulus of elasticity
UTS	Ultimate tensile strength
f	Compact Tensile Test
ϕ	beta
ϵ_p	beta
σ_y	change in yield surface size
σ_0	Initial Yield Strength of material
R	Increment of the yield stress
X	Back-stresses
p	equivalent plastic strain
c, Q, b	Material parameters

1. INTRODUCTION

Structural components in critical applications, such as nuclear reactors and pressure vessels, are often exposed to extreme cyclic loading during events like earthquakes. This loading can lead to very low cycle fatigue (VLCF), typically characterized by failure within 10 to 100 cycles. Unlike low cycle fatigue (LCF), where surface shear dominates crack

initiation, VLCF often involves internal crack nucleation due to void growth and coalescence caused by large plastic strains. These distinct mechanisms necessitate specialized evaluation models that accurately predict fatigue in life under such conditions.

Conventional LCF models may not account for the unique damage mechanisms under VLCF, particularly in multi-axial stress states. To address this, strain-based models have been developed, which incorporate parameters such as stress triaxiality and void growth. One such model, based on the Rice and Tracey void growth theory [3], evaluates fatigue life by considering both void expansion under tensile stress and shrinkage under compressive stress. This approach provides a more accurate assessment of the damage accumulation leading to crack initiation.

In this project, we implement a strain-based VLCF evaluation model and validate it through finite element (FE) simulations using the SA508 Gr.3 low alloy steel. This material is widely used in the construction of critical structural components due to its excellent mechanical properties and resistance to fracture. The simulations focus on notched compact tension (C(T)) specimens to replicate the multi-axial stress states encountered in real-world applications. By comparing the simulated fatigue life with experimental data, we aim to assess the accuracy and reliability of the strain-based model in predicting VLCF behavior in SA508 Gr.3 steel.

2. Experiments

The material properties for SA508 Gr.3, which is a low alloy steel [2] was referenced from the experiments performed by J.-M. Lee et al [1]. TABLE 1 shows its material composition.

Two physical tests were carried out on SA508 Gr. 3

- a. Ultimate tensile test
- b. Notched compact tension test under cyclic loading

Chemical Composition	Ni	Mo	Cr	V
SA508 Gr. 3	0.75	0.5	0.21	0.002

TABLE 1: SA508 Gr. 3 Material Composition

2.1 Ultimate tensile test

Tensile tests using a round bar specimen with a diameter and a gage length of 5 and 25 mm, respectively were performed by J.-M. Lee et al. [1] and stress strain properties were referred from the same. Figure 1 displays the engineering stress-strain and true stress strain curves, while Table 2 provides a summary of the tensile properties.

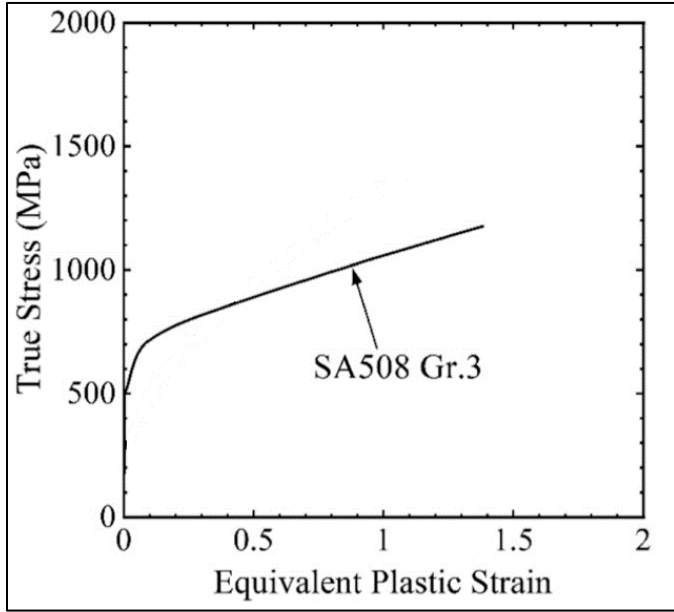


FIGURE 1: True Stress-Strain Curve from tensile test

Ultimate Tensile Strength	610 MPa
Minimum Tensile Strength	550 MPa
Initial Yield stress	498.8 MPa

TABLE 2: Tensile properties of SA508 Gr. 3

2.2 Notched compact tension test under cyclic loading

Notched compact tension (notched C(T)) tests under cyclic loading were performed to investigate crack initiation under VLCF loading conditions. The specimen geometry is depicted in FIGURE 2.

This notched C(T) test was conducted under displacement-controlled cyclic loading with the cyclic loading sequence, consisting of five sets of increasing the amplitude of the crack mouth opening displacement (CMOD), shown in FIGURE 4. Each set consisted of twenty cycles of fully reversed triangular waves with the same amplitude [1].

The crack initiation or failure cycle for notched C(T) was observed at 4th set, 15th cycle (75th cycle)

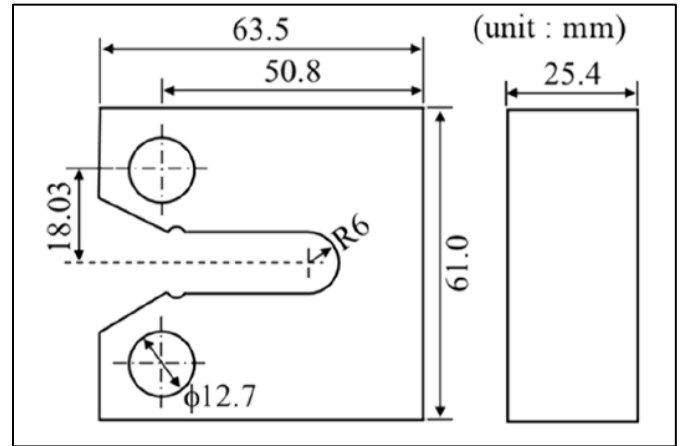


FIGURE 2: C(T) test standard coupon dimensions

3. FE ANALYSIS OF NOTCHED C(T) TEST

Elastic-plastic FE analysis was carried out to replicate the notched C(T) test using the commercial FE analysis software ABAQUS [4]. The second-order solid brick element with reduced integrations (C3D20R in ABAQUS) was employed in a three-dimensional (3-D) model, the model and mesh geometry can be seen in FIGURE 3. Throughout this investigation, a 0.5 mm element size was employed at the notch tip for computational efficiency.

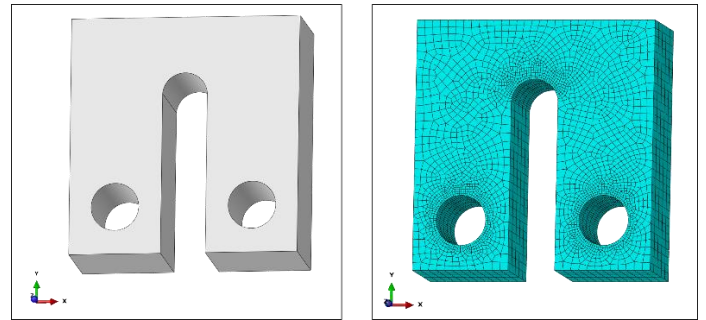


FIGURE 3: 3D model and mesh geometry from Abaqus

3.1 Boundary conditions and material properties

In the FE analysis, the displacement histories shown in FIGURE 4 that were used in the experiment are applied to the pinhole surfaces of notched C(T) specimens. The other pinhole was fixed by applying a zero-displacement boundary condition to the inner surface. According to ASME Section II Part D M, 2023 [2], the UTS, density, Poisson's ratio, and modulus of elasticity for SA508 were taken from Table U, PRD, and TM-1 tables, respectively. The elastic-plastic deformation of notched C(T) specimens under a cyclic load was simulated using ABAQUS's combined hardening model [4].

The experimental findings by J.-M. Lee et al. [1] are used to adopt the cyclic hardening properties necessary for true stress-equivalent plastic strain input. The following section goes into detail about the hardening model.

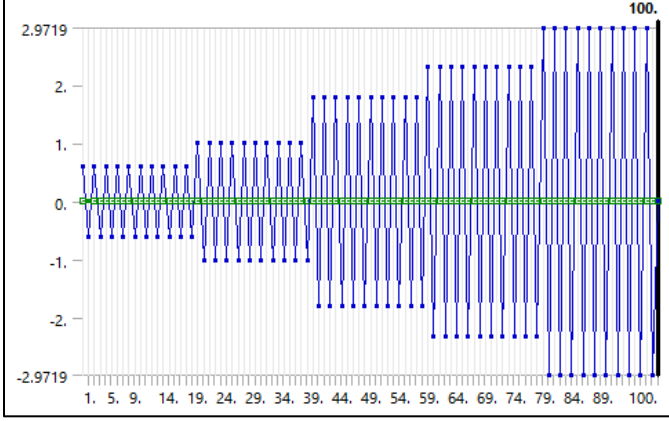


FIGURE 4: Displacement amplitude from simulating C(T) test

3.2 Cyclic Hardening Properties

For this FE simulation combined hardening model proposed by J. Chaboche et al. [5] was used. Its yield function is given by,

$$f = \varphi(\sigma - c\epsilon_p) - \sigma_y, [5]$$

$$X = c\epsilon_p$$

σ_y - Size of the yield surface

X -Back Stress

The expression for the change in yield surface size (isotropic hardening) is given as,

$$\sigma_y = \sigma_0 + R$$

$$dR = b(Q - R) dp, [5]$$

σ_0 – Initial yield strength of the material

R – Increment of the yield stress

p – equivalent plastic strain

Q, b – Material parameters

When $Q = 0$, σ_y is constant and only kinematic hardening is assumed to occur. SA508 Gr.3 does not show cyclic hardening behavior at room temperature [6]. The X is used to simulate kinematic hardening (movement of the yield surface). In the model, it is expressed in terms of p as follows,

$$X = \sum_{i=1}^m X_i$$

$$dX_i = \frac{2}{3} c_i d\epsilon_p - \gamma_i X_i dp, [5]$$

m – Number of back stresses

For the analysis three back-stresses were considered. The parameters can be traced to the true stress strain curve shown in FIGURE 5 by least square method and function in MathWorks MATLAB [7]. The stress strain curve can be traced by the function predicting the Chaboche parameters for 3 back-stresses. For this analysis the parameters from experiment were referred.

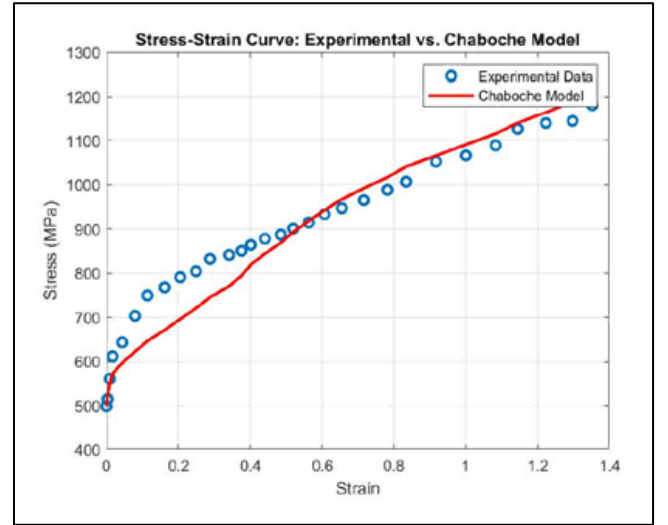


FIGURE 5: MATLAB plot for Chaboche stress-strain curve from the material parameters

4. RESULTS AND DISCUSSION

Using FE simulation results and fatigue life evaluation models described above, fatigue lives for the present notched C(T) test were evaluated. The FE analysis performed in Abaqus gave us fatigue failure at 64th cycle, as shown in TABLE 3. The stresses at the 64th cycle cross the UTS of the material can be seen in FIGURE 7. This indicated that the material undergoes necking after this point that can be referred from FIGURE 6. The UTS can be referred from TABLE 2

A compressive and tensional effect of the C(T) specimen can be observed in FIGURES 8a and 8b. This leads to an increase in cyclic stresses in the notch region causing it to fatigue failure. A comparison between experimental cycles and FE analysis cycles is shown in Table 3.

Further the Force v/s Displacement graph is extracted from the history output in Abaqus and can be seen in FIGURE 9. A loading and unloading force can be observed from the plot for the respective displacements.

Result	Experimental result	FE simulation result
SA508 Gr. 3	75th Cycle	64th Cycle (85.33%)

TABLE 3: Comparison between experimental cycles and FE analysis cycles

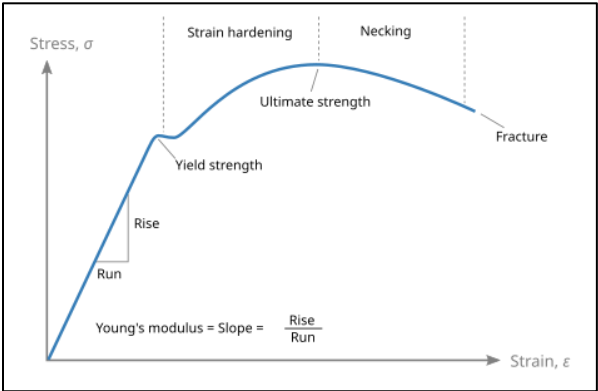


FIGURE 6: Necking explained from Stress-Strain Curve

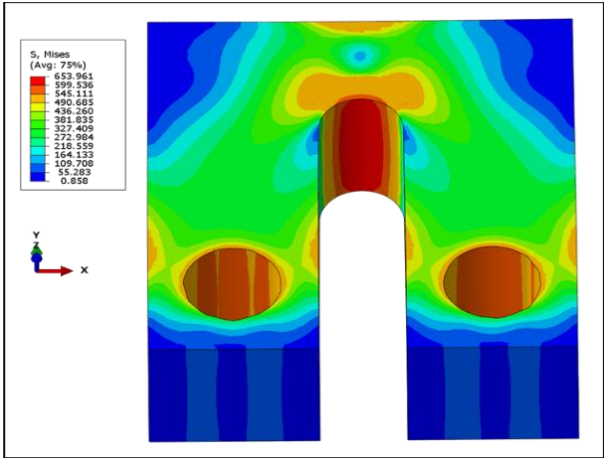


FIGURE 7: Failure cycle obtained from FE Analysis

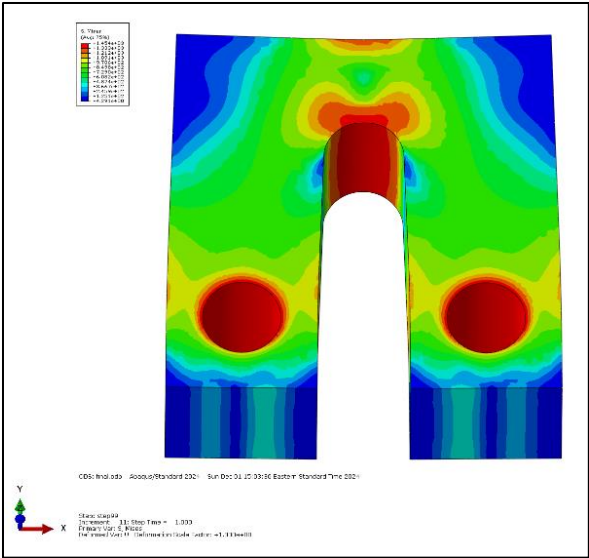


FIGURE 8a. Tension stresses visualized through FE analysis

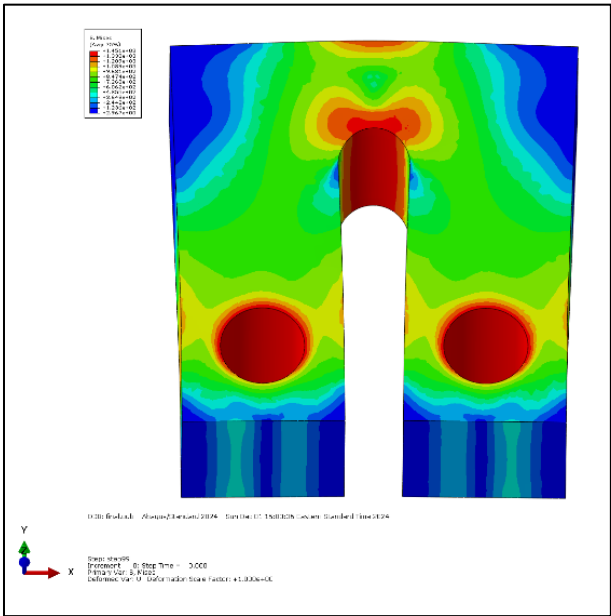


FIGURE 8b.: Compressive stresses visualized through FE analysis

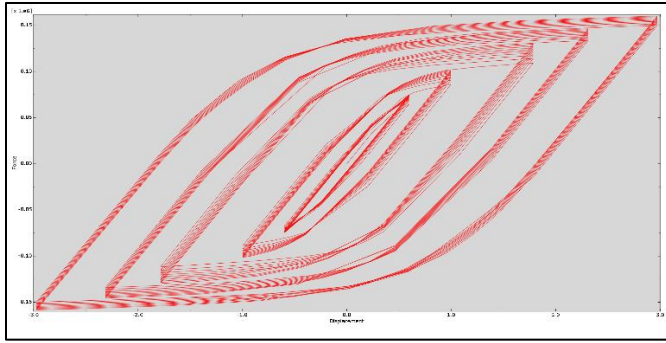


FIGURE 9: Force v/s Displacement Plot

CONCLUSION

This work validates the use of finite element analysis for predicting low-cycle fatigue behavior in SA508 Gr.3 steel. The Chaboche model was effective in simulating cyclic plasticity, and the alignment with experimental results underscores the reliability of FEA in industrial fatigue studies. The strain-based model was successfully implemented in FE software. The MATLAB function for least squares method was explored and can have a scope to be perfectly implemented to find the Chaboche parameters. A detailed linear regression should also be considered for future scope to find these parameters.

The insights gained from this study reinforce the importance of LCF analysis in designing durable components for high-stress environments, such as pressure vessels, turbine blades, and structural steel applications in earthquake-prone areas.

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